

WHY BIT MANIPULATION?

Bitwise operations manipulate individual bits directly inside CPU registers in $O(1)$ time with 0 extra memory! They power low-level systems, cryptographic algorithms, and optimal interview solutions.

THE KERNIGHAN'S TRICK
AHA

$n \& (n - 1)$ clears the lowest set bit (1) of n in exact $O(1)$ time! Use this to count set bits in $O(\text{set bits})$ instead of 32 loop iterations!

6 ESSENTIAL BIT OPERATORS

- AND (&):** Both bits 1 $\rightarrow$ 1. Masking.
- OR (|):** Either bit 1 $\rightarrow$ 1. Setting bits.
- XOR (^):** Different bits $\rightarrow$ 1. $x \oplus x = 0, x \oplus 0 = x$!
- NOT (~):** Bitwise inversion.
- Left Shift (<<):** Multiply by 2^k .
- Unsigned Right Shift (>>):** Fills 0 on left.

2 VISUALIZING KERNIGHAN'S TRICK

Clearing lowest set bit of 12 (1100):

```
n      = 12 -> 1100 (binary)
n - 1  = 11  -> 1011 (binary)
n & (n - 1) = 1000 (binary) -> 8!
Lowest set bit at position 2 is CLEARED!
```

PREREQUISITES & COMPLEXITY REASONING

Operation / Problem	Time Complexity	Auxiliary Space
Single Number (XOR Accumulation)	$O(N)$ single pass	$O(1)$ space
Hamming Weight ($n \& (n-1)$)	$O(\text{set bits}) \leq 32$	$O(1)$ space
Counting Bits (DP + Bitshift)	$O(N)$ single pass	$O(N)$ result array
Sum of 2 Integers (XOR + AND Carry)	$O(32) = O(1)$	$O(1)$ space

CLUES THAT SCREAM BIT MANIPULATION

- "Every element appears twice except one" $\rightarrow$ Accumulative XOR ($a \oplus \text{num}$)
- "Count total 1 bits in 32-bit integer" $\rightarrow$ Kernighan's $n = n \& (n - 1)$
- "Add two numbers without + or -" $\rightarrow$ Bitwise XOR ($a \oplus b$) + Carry ($(a \& b) \ll 1$)

1 5 MUST-KNOW BIT TRICKS

```
// 1. Get k-th bit (0 or 1):
int bit = (n >> k) & 1;

// 2. Set k-th bit to 1:
n = n | (1 << k);

// 3. Clear k-th bit to 0:
n = n & ~(1 << k);

// 4. Toggle k-th bit:
n = n ^ (1 << k);

// 5. Check if n is power of 2:
boolean isPowerOfTwo = (n > 0) && ((n & (n - 1)) == 0);
```

2 KERNIGHAN'S BIT COUNTING HELPER

```
public static int hammingWeight(int n) {
    int count = 0;
    while (n != 0) {
        n = n & (n - 1); // Clears lowest set bit!
        count++;
    }
    return count;
}
```

Loops ONLY for set bits ('1'), skipping all 0 bits!

PATTERN 1 SINGLE NUMBER (XOR Self-Cancellation) e.g. Single Number (LC 136)	
XOR CANCELLATION MATH \$x \oplus x = 0\$ and \$x \oplus 0 = x\$. Since all duplicate pairs cancel to 0, XORing all array numbers leaves ONLY the unique single number!	TEMPLATE — LC 136 <pre>public int singleNumber(int[] nums) { int res = 0; for (int num : nums) { res ^= num; // Pair cancellation! } return res; }</pre>
PATTERN 2 MISSING NUMBER (Index XOR Array Accumulation) e.g. Missing Number (LC 268)	
INDEX MATCHING STRATEGY XOR all indices 0..N with all array values `nums[i]`. Every present number cancels its matching index, leaving ONLY the missing number!	TEMPLATE — LC 268 <pre>public int missingNumber(int[] nums) { int res = nums.length; for (int i = 0; i < nums.length; i++) { res ^= i ^ nums[i]; } return res; }</pre>

PATTERN 3

COUNTING BITS (DP Bitshift Recurrence) e.g. Counting Bits (LC 338)

RECURRENCE FORMULA `dp[i] = dp[i >> 1] + (i & 1)`. `i >> 1` shifts off the last bit, which was computed earlier! O(N) time.	TEMPLATE — LC 338 <pre>public int[] countBits(int n) { int[] res = new int[n + 1]; for (int i = 1; i <= n; i++) { res[i] = res[i >> 1] + (i & 1); } return res; }</pre>
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PATTERN 4

REVERSE BITS (32-Bit Reversal Accumulator) e.g. Reverse Bits (LC 190)

BIT SHIFT ACCUMULATOR Loop 32 times: shift `res` left by 1, OR with lowest bit `n & 1`, shift `n` right by 1 (`n >>= 1`).	TEMPLATE — LC 190 <pre>public int reverseBits(int n) { int res = 0; for (int i = 0; i < 32; i++) { res = (res << 1) (n & 1); n >>= 1; // Unsigned right shift! } return res; }</pre>
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PATTERN 5

SUM OF TWO INTEGERS (XOR Sum + AND Carry)

e.g. Sum of Two Integers (LC 371)

HALF-ADDER HARDWARE LOGIC

`a ^ b` computes addition WITHOUT carry.
`(a & b) << 1` computes carry.
Repeat until carry `b == 0`!

TEMPLATE — LC 371

```
public int getSum(int a, int b) {  
    while (b != 0) {  
        int carry = (a & b) << 1;  
        a = a ^ b;  
        b = carry;  
    }  
    return a;  
}
```

PATTERN 6

REVERSE INTEGER (32-Bit Overflow Guard) e.g. Reverse Integer (LC 7)

OVERFLOW GUARD FORMULA

Check ``res > Integer.MAX_VALUE / 10`` or ``res < Integer.MIN_VALUE / 10`` before multiplying by 10!

TEMPLATE — LC 7

```
public int reverse(int x) {
    int res = 0;
    while (x != 0) {
        int pop = x % 10; x /= 10;
        if (res > Integer.MAX_VALUE / 10 || (res == Integer.MAX_VALUE / 10 && pop > 7))
            return 0;
        if (res < Integer.MIN_VALUE / 10 || (res == Integer.MIN_VALUE / 10 && pop < -8))
            return 0;
        res = res * 10 + pop;
    }
    return res;
}
```

BIT MANIPULATION — WHICH TRICK TO USE?

Start: What is the bitwise goal?

Find single non-duplicate number

→

Accumulative XOR

``res ^= num``
(pairs cancel)

Count total 1 bits (Hamming weight)

→

Kernighan's Trick

``n = n & (n - 1)``

Count bits for all 0..N numbers

→

DP Bitshift

``dp[i] = dp[i >> 1] + (i & 1)``

+ Add numbers without + or -

→

XOR + AND Carry

``a ^ b``
AND ``a & b``
Carry ``a & b << 1``

TRIGGER WORDS DIRECTORY

Trigger Word	Variant	Action
"Single number / unique element"	XOR Accumulation	<code>`res ^= num`</code>
"Number of 1 bits"	Kernighan's	<code>`n = n & (n - 1)`</code>
"Count set bits for 0..N"	DP Bitshift	<code>`res[i] = res[i >> 1] + (i & 1)`</code>
"Sum without + operator"	Bitwise Addition	Loop <code>`a = a ^ b`</code> , <code>`b = carry`</code>

INTERVIEW TRADE-OFFS & FOLLOW-UPS

Interviewer Asks	Your Answer
Why <code>`>>>`</code> instead of <code>`>>`</code> in Reverse Bits?	<code>`>>>`</code> is unsigned right-shift (fills 0 on left). <code>`>>`</code> preserves sign bit (fills 1 for negative numbers)!
How to check if N is power of 2?	<code>`(n > 0) && ((n & (n - 1)) == 0)`</code> . Powers of 2 have exactly ONE set bit!

EASY — XOR & Bit Counting

LC 136 · Single Number EASY

Pattern: XOR Self-Cancel
Key Insight: `res ^= num`.

```
public int singleNumber(int[] nums) {
    int res = 0;
    for (int n : nums) res ^= n;
    return res;
}
```

LC 191 · Number of 1 Bits EASY

Pattern: Kernighan's Trick
Key Insight: `n = n & (n - 1)`.

```
public int hammingWeight(int n) {
    int c = 0;
    while (n != 0) { n &= (n - 1); c++; }
    return c;
}
```

LC 268 · Missing Number EASY

Pattern: Index XOR Match
Key Insight: `res ^= i ^ nums[i]`.

```
public int missingNumber(int[] nums) {
    int res = nums.length;
    for (int i = 0; i < nums.length; i++) res ^= i ^ nums[i];
    return res;
}
```

EASY & MEDIUM — Counting Bits & Reverse

LC 338 · Counting Bits EASY

Pattern: DP Bitshift
Key Insight: `dp[i] = dp[i >> 1] + (i & 1)`.

```
public int[] countBits(int n) {
    int[] res = new int[n + 1];
    for (int i = 1; i <= n; i++) res[i] = res[i >> 1] + (i & 1);
    return res;
}
```

LC 190 · Reverse Bits EASY

Pattern: 32-Bit Unsigned Shift
Key Insight: `res = (res << 1) | (n & 1); n >>= 1`.

```
public int reverseBits(int n) {
    int res = 0;
    for (int i = 0; i < 32; i++) {
        res = (res << 1) | (n & 1); n >>= 1;
    }
    return res;
}
```


MEDIUM — Bitwise Addition & Overflow Reverse

LC 371 · Sum of Two IntegersMEDIUM

Pattern: XOR + AND Carry

Key Insight: ``a = a ^ b`, `b = (a & b) << 1`.`

```
public int getSum(int a, int b) {
    while (b != 0) {
        int carry = (a & b) << 1;
        a = a ^ b; b = carry;
    }
    return a;
}
```

MEDIUM — Integer Reversal

LC 7 · Reverse IntegerMEDIUM

Pattern: Overflow Protection

Key Insight: Guard ``res > MAX / 10`` before multiplying.

```
public int reverse(int x) {
    int res = 0;
    while (x != 0) {
        int pop = x % 10; x /= 10;
        if (res > Integer.MAX_VALUE / 10 || res < Integer.MIN_VALUE / 10) return 0;
        res = res * 10 + pop;
    }
    return res;
}
```

COMPLETE BIT MANIPULATION PROBLEM LADDER SUMMARY					
LC #	Problem	Pattern	Time	Space	Diff
136	Single Number	XOR Self-Cancellation	O(N)	O(1)	E
191	Number of 1 Bits	Kernighan's Trick	O(set bits)	O(1)	E
268	Missing Number	Index XOR Match	O(N)	O(1)	E
338	Counting Bits	DP Bitshift Recurrence	O(N)	O(N)	E
190	Reverse Bits	32-Bit Unsigned Shift	O(32)	O(1)	E
371	Sum of Two Integers	XOR + AND Carry	O(32)	O(1)	M
7	Reverse Integer	Overflow Protection	O(log10 X)	O(1)	M

🔍 DRY RUN — Single Number ([4, 1, 2, 1, 2])

Step	num	XOR Operation ('res ^= num')	res Value
1	4	<code>'0 ^ 4'</code>	4
2	1	<code>'4 ^ 1'</code>	5
3	2	<code>'5 ^ 2'</code>	7
4	1	<code>'7 ^ 1'</code> (cancels previous 1)	6
5	2	<code>'6 ^ 2'</code> (cancels previous 2)	4

📐 MATHEMATICAL PROOF — KERNIGHAN'S CLEAR BIT FORMULA

Claim: `'n & (n - 1)'` clears the lowest set bit of `'n'`.

Proof: Subtracting 1 flips the rightmost `'1'` to `'0'` and all lower trailing `'0's` to `'1's`. Performing `'n & (n - 1)'` keeps all higher bits unchanged while forcing the rightmost `'1'` to `'0'` $\rightarrow$ **Exact 1 bit cleared!**

✅ MASTERY CHECKLIST — CAN YOU DO THIS?

☐ Code Single Number in 1m with XOR

☐ Write Kernighan's `'n & (n - 1)'` bit counter

☐ Code Missing Number with index XOR

☐ Write Counting Bits with DP bitshift

☐ Code Reverse Bits using `'>>>'` shift

☐ Write Sum of Two Integers (XOR + Carry)

☐ Write Reverse Integer with overflow guard

☐ Prove why `'n & (n - 1) == 0'` tests power of 2

📁 GOLDEN RULES — NEVER FORGET

Rule 1 — Operator Precedence Warning
Bitwise operators (`'&'`, `'|'`, `'^'`) have LOWER precedence than comparison operators (`'=='`, `'<'`)! ALWAYS wrap bitwise expressions in parentheses: `'((n & 1) == 0)'`!

Rule 2 — Always Use Unsigned Right Shift for Reversal
In 32-bit bit manipulation, use `'>>>'` instead of `'>>'` so negative numbers do not fill leading 1s!